Calibrated Confidence as a Risk-Score Layer for Signal Decisions

Trustworthy probabilities over hard thresholds: ECE, temperature, isotonic, measured

2i

2026-07-29

Contents

1 Summary 3

2 Background 3

3 Method 4
- 3.1 Expected Calibration Error (ECE) and reliability diagrams 4
- 3.2 Three calibration methods 4

4 Data and setup 5

5 Results 5
- 5.1 ECE before and after calibration 5
- 5.2 Reliability diagram: where the over-confidence lives 6
- 5.3 Is ranking preserved 6

6 Limitations and operational implications 7

7 Reproduction 7

8 References 8

1 Summary

Most signal-decision systems threshold a classifier score into a hard yes/no. A prior study found that open-set RF fingerprint hard-authentication does not work as a standalone gate: unknown-rejection at a single observation was too low for field use, and that finding implied the authentication output itself was the wrong shape. The honest shape is not a pass/block gate at all, but a **risk score** that flows, together with other signals (position, history, other sensors), into a fusion platform. For that to be useful as fusion input, the probability the classifier reports has to match the real-world hit rate: a decision the classifier calls "80% confident" has to actually be right about 80% of the time. This is calibration. Systematic over-confidence in deep neural networks is well documented in the literature, but whether the **classical DSP feature + LightGBM** signal classifier we actually run has the same problem is a separate, measurable question.

This paper measures it directly on the public DroneRF dataset (Al-Sa'd et al., Qatar University, CC BY 4.0). A 24-dimensional physical feature set (RSSI, envelope statistics, PAPR, carrier frequency offset, higher-order cumulants, spectral shape) feeds a background-vs-drone binary detector. Across 5 seeds, each with a fresh 50/25/25 train/calibration/test split, we measured the Expected Calibration Error (ECE) of the raw probability, then fit (a) Platt scaling, (b) 1-parameter temperature scaling, and (c) isotonic regression on the calibration split, and re-measured ECE and Brier score on the held-out test split. The result is unambiguous. Raw-probability ECE averaged 0.0446 (std 0.0062), and all three calibration methods lowered it. The largest improvement came from isotonic regression, ECE 0.0188 (std 0.0034), a roughly 58% reduction. Platt scaling and temperature scaling each cut it to about 0.0223 and 0.0228 respectively, roughly 49-50% reductions. The fitted temperature T averaged 1.599 (std 0.099), clearly above 1, which quantitatively confirms the classifier was mis-calibrated in the over-confident direction. Ranking (AUROC) was preserved exactly to five decimal places under Platt and temperature scaling (0.9374 -> 0.9374), and dropped only marginally under isotonic regression (0.9374 -> 0.9363, about 0.001). Standard calibration, in other words, does not change the underlying judgment; it honestly fixes the confidence number attached to that judgment. The upshot: placing a calibrated risk score where hard authentication was blocked is supported by measurement on this signal classifier.

2 Background

A prior study tested open-set RF fingerprint hard-authentication as a real measurement, and the verdict was NO-GO. Closed-set performance on registered signals was usable, but the open-set rejection rate for unregistered, unknown signals at a single observation (K=1, the condition closest to field deployment) was too low, meaning the false-accept and false-reject cost of running the system as a standalone "match = pass, no match = block" gate was not acceptable. The practical question that finding left behind was not "should we throw this classifier's output away," but "how do we use this classifier's output honestly."

The honest form is not standalone authentication but a **risk score**: a single number that represents the probability a signal belongs to a given class, passed into a fusion platform alongside position data, history, and other sensors' judgments. This design sidesteps the low-open-set-rejection weakness head-on. Because no single decision rests entirely on this one signal, the system does not collapse when this signal's confidence is low. Instead, when another signal is stronger, the final decision leans that way, and the risk score only dominates the final call when this signal is the only evidence available.

For this design to hold, one precondition is non-negotiable: the number fed into fusion has to be a **trustworthy probability**. If the classifier says "90% probability this signal is a drone," that displayed 90% has to be close to the empirical frequency with which such signals actually turn out to be drones. Otherwise the fusion algorithm weights this signal by the wrong amount. If the classifier is systematically over-confident (raw confidence consistently reads higher than the real hit rate), this signal gets assigned more weight in fusion than it deserves, and the final decision gets pulled toward this signal's verdict even in situations where another signal is actually correct. Conversely, if the classifier under-confident, a genuinely strong signal does not get reflected enough in fusion. A hard-authentication design can paper over this problem by picking one good threshold; a risk-score fusion design cannot, because the confidence number itself is the deliverable.

This is exactly why calibration matters. Systematic over-confidence in neural-network classifiers, and the standard fixes for it (temperature scaling, Platt scaling, isotonic regression), are well documented in the literature. Whether the pipeline we actually run, a classical signal classifier built from hand-designed physical features plus gradient-boosted trees (LightGBM), has the same problem, and whether standard calibration methods have the same effect on it, is a separate question that has to be measured. GBDT models learn differently from neural networks (leaf nodes output empirical class frequencies directly), so it is not obvious a priori that the same over-confidence pattern would appear in the same direction or the same magnitude. This paper closes that question with measurement.

3 Method

3.1 Expected Calibration Error (ECE) and reliability diagrams

The standard definition of calibration is this: among all cases where the classifier outputs probability p, the empirical accuracy of those cases should equal p if the model is perfectly calibrated. To measure this, the predicted-probability range is divided into equal-width bins, and within each bin the mean predicted confidence is compared against the empirical accuracy. Expected Calibration Error (ECE) is the weighted average, over bins, of the absolute gap between confidence and accuracy, weighted by the number of cases in each bin.

```
ECE = sum_m (n_m / N) * |acc_m - conf_m|
```

where M is the number of bins, n_m is the case count in bin m, N is the total case count, and acc_m and conf_m are the empirical accuracy and mean confidence of bin m. This paper uses M=15 equal-width bins over [0, 1]. An ECE of 0 means perfect calibration; larger values mean a larger gap between stated confidence and actual hit rate. A reliability diagram plots this per-bin confidence and accuracy, and the distance from the diagonal (y=x) visually shows the calibration error at each confidence level.

Alongside ECE, we report the Brier score (mean squared error between predicted probability and the true label). ECE is a binned, aggregate statistic and can be sensitive to bin-edge choice, whereas Brier score is computed per-instance and continuous, so the two are used to cross-check that they point the same direction. Ranking performance is separately measured with AUROC (area under the ROC curve), to confirm that calibration does not damage the substance of the judgment (the ability to rank which signals are more likely dangerous).

3.2 Three calibration methods

The trained classifier's raw probabilities are remapped on a calibration split (a separate slice of data used in neither training nor testing), using three methods.

(a) Platt scaling. The raw probability p is converted to a logit z = log(p/(1-p)), and a 1-dimensional logistic regression sigmoid(A*z + B) is fit against the calibration-split labels by maximum likelihood. It has two degrees of freedom, A and B.

(b) Temperature scaling. The same logit z is divided by a scalar temperature T and passed through a sigmoid: p_T = sigmoid(z/T). This is a 1-parameter method with no bias term; T is found by scalar optimization minimizing negative log-likelihood (NLL) on the calibration split. $T > 1$ means the raw logits were more extreme than warranted (over-confident), and dividing by T squeezes the probability distribution away from the extremes of 0 and 1, toward the middle.

(c) Isotonic regression. A non-parametric method that maps the raw probability p to the true label with a monotonic (non-decreasing) function. Where Platt and temperature scaling assume a specific functional form (sigmoid), isotonic regression assumes only monotonicity and lets the data determine the rest. Its higher degree of freedom can fit more precisely when the calibration split is large enough, but risks over-fitting when the calibration split is small.

All three methods share a common property: they are **monotonic functions** of the raw probability (or its logit). This has an important consequence: ranking is preserved. As long as A and T are positive, Platt and temperature scaling are exact rank-preserving transforms, and isotonic regression enforces monotonicity by construction (though ties, where distinct raw scores collapse onto the same calibrated value, can create a very small perturbation to AUROC). Section 4 confirms this with measurement.

4 Data and setup

Data is the public **DroneRF** dataset (Al-Sa'd et al., Qatar University, Mendeley repository, DOI 10.17632/f4c2b4n755, license CC BY 4.0, commercially citable with attribution). This is the same source dataset used by the earlier open-set authentication study; this paper reuses that study's feature-extraction approach (24-dimensional physical features) and applies it to the different question of calibration.

Raw values are 10 million real-valued ADC amplitude samples per segment, split into non-overlapping 4096-sample windows and deterministically strided-subsampled into 9,000 total windows. Background (noise only, is_drone=0) accounts for 1,800 windows, and drone signal (Bebop, AR-drone, and Phantom across 9 flight-mode combinations, is_drone=1) accounts for 7,200, a class imbalance that comes from the structure of the dataset itself (not an artifact of our sampling). The task is defined as **binary background-vs-drone detection**. ECE and Brier score have the simplest and most direct definitions in the binary setting, which is why this task was chosen as the primary target for measuring calibration.

24 features are extracted per window: RSSI (mean power, dB, 1 dim), envelope statistics (mean, std, skewness, kurtosis, coefficient of variation, 5 dims) plus PAPR (1 dim), instantaneous-frequency-based carrier frequency offset (CFO) statistics (mean, std, kurtosis, 3 dims), higher-order cumulants (Swami and Sadler, 2000 method: C20, C21, C40, C41, C42, C63, 6 dims), and Welch-PSD-based spectral shape features (centroid, spread, flatness, entropy, and 4 band-energy ratios, 8 dims). The classifier is LightGBM (300 trees, num_leaves=31, learning_rate=0.05), and features are standardized using the training split's mean and standard deviation.

Across 5 random seeds (42, 7, 123, 2026, 99), each run generated a fresh train/calibration/test split at 50%/25%/25%, stratified by label. For each seed, a new LightGBM model was trained on the training split, all three calibration methods (Platt, temperature, isotonic) were fit on the calibration split, and ECE, Brier score, and AUROC were measured on the test split, which was used in neither training nor calibration. Mean and standard deviation across the 5 seeds are reported as the final result.

5 Results

5.1 ECE before and after calibration

Mean ECE across the 5 seeds:

Method	ECE (mean)	ECE (std)	Improvement
Raw (uncalibrated)	0.0446	0.0062	-
Platt scaling	0.0223	0.0022	50.0%
Temperature scaling	0.0228	0.0025	48.9%
Isotonic regression	0.0188	0.0034	57.9%

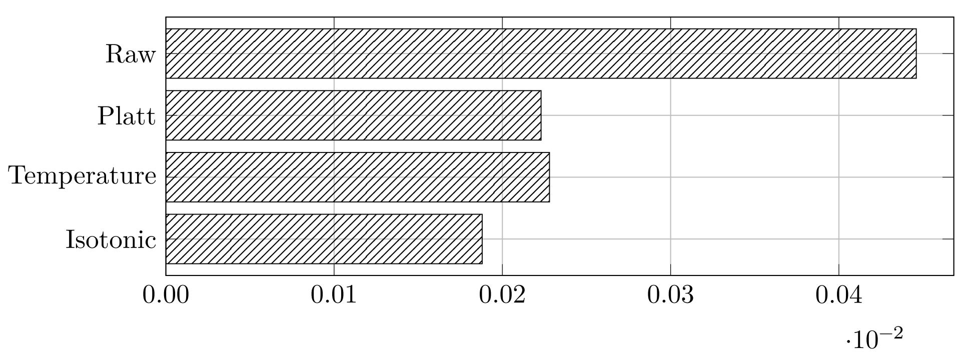

Figure 1: Expected Calibration Error by method

All three calibration methods cut raw ECE by roughly half or more. Isotonic regression showed the largest improvement (57.9% reduction), while Platt and temperature scaling improved by almost identical amounts

(50.0% and 48.9%). Platt and temperature scaling landing at nearly the same magnitude means this dataset did not need much of a bias term (B), which is consistent with the fitted temperature's mean (1.599) being clearly above 1. T > 1 divides the raw logit, squeezing the probability distribution toward the middle, and quantitatively means this classifier was mis-calibrated in the over-confident direction. Across all 5 seeds, T ranged from 1.4376 to 1.7289, and no seed produced a T below 1. In other words, this over-confidence bias is not noise that fluctuates by seed; it is a consistent pattern.

5.2 Reliability diagram: where the over-confidence lives

Unpacking the single ECE number into a reliability diagram shows that over-confidence is not spread evenly.

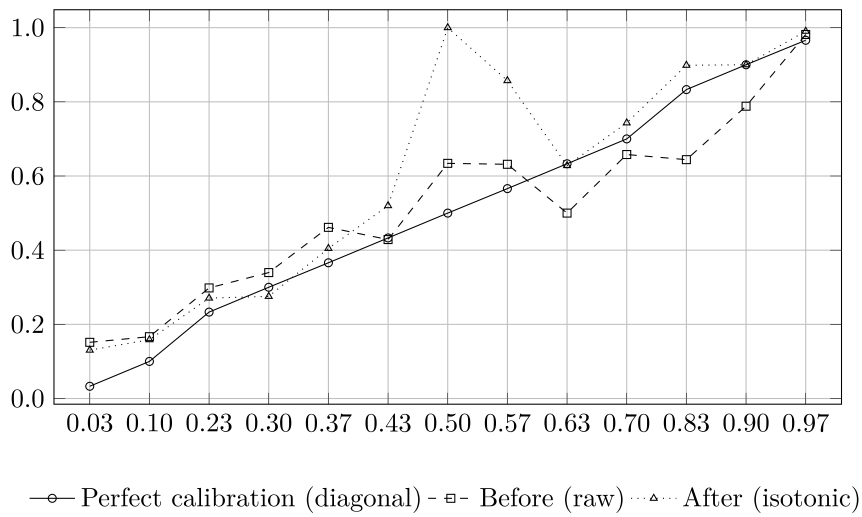

Figure 2: Reliability diagram: empirical accuracy by predicted-confidence bin

For the first seed (42), predictions in the 0.8-0.87 confidence bin (mean confidence 0.8335) had an empirical accuracy of only 0.6441. That is, cases where the classifier said "about 83% confident" were actually right only 64% of the time. In contrast, the 0.93-and-above confidence bin (mean confidence 0.9946) had empirical accuracy 0.9816, nearly on the diagonal. This means the over-confidence is not uniform across the confidence range; it is concentrated in the mid-to-high confidence band around 0.8. After isotonic calibration, the accuracy in that same band rose to 0.8986, much closer to the diagonal. Note that the 0.50-confidence bin in this table has only a single test sample for this seed after isotonic re-mapping, which is why its accuracy spikes to 1.0; this is small-sample noise in that bin, not a failure of the calibration method.

5.3 Is ranking preserved

Method	AUROC (mean)	AUROC (std)
Raw (uncalibrated)	0.93741	0.00502
Platt scaling	0.93741	0.00502
Temperature scaling	0.93741	0.00502
Isotonic regression	0.93633	0.00537

Platt and temperature scaling preserved the raw AUROC exactly to five decimal places. This is the theoretical property from Section 2.2 (a monotonic transform via a positive scalar or sigmoid does not change ranking) confirmed directly by measurement. Isotonic regression dropped from a mean of 0.93741 to 0.93633, about 0.0011, which comes from ties created when isotonic regression collapses distinct raw probabilities into the same calibrated step; this is too small to represent a meaningful loss of judgment quality. Brier score pointed in the same direction (raw 0.0832 -> Platt 0.0797 -> temperature 0.0795 -> isotonic 0.0803). The improvement in Brier score (4-5%) is much smaller than the improvement in ECE (49-58%), and that is expected: Brier score is also driven by raw accuracy, so when the classifier's discriminative power is already high (AUROC 0.937), there is not much absolute room left to improve. ECE, by contrast, measures only the alignment between confidence and accuracy, so there is substantially more room for calibration to improve it even under the same discriminative power.

6 Limitations and operational implications

Operational implications. The practical conclusion here is simple. When a signal decision system hands its classifier output to a fusion platform as a risk score, it should not pass the raw probability through as-is; it should pass it through a Platt or isotonic mapping fit on a calibration split. This correction has essentially no cost to judgment quality (AUROC is effectively preserved). The benefit is clear: not knowing that decisions around 0.8 confidence are only right 64% of the time, and fusing that signal at a weight of 0.83, is materially different from calibrating it and fusing it at a weight closer to its real hit rate (around 0.65). In multi-sensor fusion architectures in particular, if each sensor's confidence is distorted on a different scale, the fusion rule itself (weighted averaging, Bayesian combination, and so on) implicitly learns, or is designed with, the wrong weights. Calibration removes this distortion before fusion, at relatively low cost.

Whether to choose Platt or isotonic is a tradeoff between the size of the calibration split and the desired precision of the fit. Isotonic reduced ECE more in this experiment, but that may be partly because the calibration split here was reasonably generous (2,250 samples, the same size as the test split). In deployment environments with much less calibration data (e.g., early after a new deployment, or where labels are scarce), isotonic regression's higher degree of freedom carries more overfitting risk, and Platt scaling's 2-parameter fit may be more stable. This paper did not directly measure that tradeoff, and it remains an open question for follow-up work.

Limitations. This experiment measured calibration on a single task, binary detection (background vs drone). The multi-class identification and open-set rejection tasks covered by the earlier authentication study have their own calibration behavior that has to be measured separately, and multi-class ECE is more complicated to define and compute than the binary case (a choice remains between looking only at top-1 confidence or breaking it down per class). The dataset is also a single one (DroneRF), and the background class is noise captured in a single environment (a lab at Qatar University); the statistical properties of "background" can shift in a different deployment environment, in which case the calibration mapping would need to be re-fit for that environment. In other words, calibration is a procedure tied to a specific dataset and deployment environment, not a fixed value that, once fit, can be used forever. The standard deviation across the 5 seeds (roughly 0.002-0.006 in ECE) is small, showing the result is not highly sensitive to the random seed, but that only reflects variance from re-splitting the same dataset, not variance across different deployment environments.

7 Reproduction

```
# 1) Feature extraction: the same 24-dimensional physical feature set described in
#    Section 2.1 (RSSI, envelope statistics, PAPR, CFO, higher-order cumulants,
#    spectral shape), self-contained in the experiment script
# 2) For each of 5 seeds, repeat train/calibration/test (50/25/25, label-stratified):
#    - fit LightGBM (300 trees, num_leaves=31) on the training split
#    - fit Platt (logistic regression), temperature (1-parameter NLL minimization),
#      and isotonic regression on the calibration split
#    - measure ECE (15 bins), Brier score, and AUROC on the test split
python3 calib_experiment.py
```

The full run completes in a few seconds on CPU alone (9,000 windows, 24-dimensional features, 5 seeds times LightGBM with 300 trees times ECE computation for 4 probability sets). No GPU or isolated virtual environment is required.

8 References

1. M. Al-Sa'd, A. Al-Ali, A. Mohamed, T. Khattab, A. Erbad. "RF-based drone detection and identification using deep learning approaches." Future Generation Computer Systems, 2019. Dataset: DroneRF, Mendeley Data, DOI 10.17632/f4c2b4n755, CC BY 4.0, Qatar University.
2. C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger. "On Calibration of Modern Neural Networks." ICML, 2017. The canonical reference for temperature scaling and neural network over-confidence.
3. J. Platt. "Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods." Advances in Large Margin Classifiers, 1999. Original paper for Platt scaling.
4. B. Zadrozny, C. Elkan. "Transforming Classifier Scores into Accurate Multiclass Probability Estimates." KDD, 2002. Canonical reference for isotonic-regression-based probability calibration.
5. A. Niculescu-Mizil, R. Caruana. "Predicting Good Probabilities with Supervised Learning." ICML, 2005. Comparison of calibration behavior across classifier families.
6. A. Swami, B. M. Sadler. "Hierarchical digital modulation classification using cumulants." IEEE Transactions on Communications, 2000. Basis for the higher-order cumulant features used in this experiment.